N-F-170b
September 10, 1964

SUPERSEDING
Int. Fed. Spec. N-F-00170a (AGR-ARS)
August 15, 1962 and
Fed. Spec. N-F-170
May 4, 1951

FEDERAL SPECIFICATION

FEED, DRY, FOR DOGS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for the preparation and packaging of dry feed for dogs.

1.2 Classification.

1.2.1 *Classes and forms.* Dry feed for dogs covered by this specification shall be of the following classes and forms, as specified (see 6.1):

Dog food with cooked cereals.
- Class 1—Low fat.
- Class 2—High fat.
- Class 3—Low fat and high protein.
- Form A—Meal
- Form B—Compressed pellets.
- Form C—Expanded pellets.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and Standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification:

Federal Specifications:

N-F-212—Feedstuffs, Concentrated.
SS-S-31—Salt, Table and Sodium Chloride Tablets.
UU-S-48—Sacks, Shipping, Paper.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the

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procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

Governmental:

Federal Food, Drug, and Cosmetic Act and General Regulations for Its Enforcement.

Handbook of Official Grain Standards of the United States.

(Copies may be obtained upon application, accompanied by money order, coupon, or cash to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

Regulations Governing the Meat Inspection of the United States Department of Agriculture.

(Application for copies should be addressed to the Meat Inspection Division, Agricultural Research Service, U.S. Department of Agriculture, Washington, D.C. 20250.)

Nongovernmental:

Official Methods of Analysis of the Association of Official Agricultural Chemists.

(Official Methods of Analysis of the Association of Official Agricultural Chemists is published by the Association of Official Agricultural Chemists, P. O. Box 540, Benjamin Franklin Station, Washington, D.C. 20044.)

U. S. Pharmacopeial Convention Publication:

The Pharmacopeia of the United States.

(Copies are available from Mack Publishing Company, Easton, Pennsylvania).

3. REQUIREMENTS

3.1 **Preaward sample.** When specified (see 6.1), a sample of the finished product shall be submitted to the contracting officer for approval before production is commenced.

3.2 **Components.** Components shall comply with the requirements of table I. The ingredients shall be as fresh as possible, but no ingredient (except iron oxide) shall be more than 18 months old at the time of manufacture. Tankage or components containing dried blood, hair, leather meal or hoof meal shall not be used.

TABLE I—*Components*

Component	Specification or grade requirement
1. Meat meal 55% min. protein	N-F-212
2. Meat and bone meal	N-F-212
3. Fish meal, 60% min. protein	N-F-212
5. Skimmed milk, dried	N-F-212
6. Soybean meal	N-F-212
7. Defatted wheat germ meal	N-F-212
8. Wheat middlings	N-F-212
9. Steamed bone meal	N-F-212
10. Brewer's dried yeast	N-F-212
11. Alfalfa leaf meal	N-F-212
12. Vitamin A & D oil	N-F-212
13. Fried fermentation solubles	N-F-212
14. Oat, groats	N-F-212
15. Barley, hulled	N-F-212
16. Salt, iodized	SS-S-31
17. Corn, dent	No. 2[1] yellow
18. Cereals, processed wheat	No. 2[1] or better of the classes hard red winter or hard red spring.
19. Animal fat.	Iodine value 48-60 and antioxident as permitted by USDA.[2]
20. Iron oxide (ferric oxide)	As permitted by FDA.[3]
21. Corn germ meal	None applicable
22. Wheat flakes	None applicable
23. Corn flakes	None applicable
24. Fish liver and glandular meal	None applicable
25. Vitamin A feeding oil	None applicable
26. Riboflavin supplement	None applicable
27. Certified food color	None applicable
28. D-activated plant sterol	None applicable
29. Meat by-product	None applicable
30. Flavor additive	None applicable

[1]See Handbook of Official Grain Standards of the United States.

[2]See the Regulations Governing the Meat Inspection of the United States Department of Agriculture.

[3]See the Federal Food, Drug, and Cosmetic Act and General Regulations for Its Enforcement.

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TABLE II—*Ingredients*

Component	Class 1	Class 2	Class 3
Meat meal, 55% minimum protein	X	X	
Meat and bone meal			X
Fish meal, 60% minimum protein	X	X	
Skimmed milk, dried	X[1]	X[1]	X
Corn germ meal			X
Soybean meal	X	X	X
Defatted wheat germ meal	X	X	X
Wheat flakes			X
Wheat flour middlings			X
Fish meal			X
Fish liver and glandular meal			X
Steamed bone meal	X	X	
Animal fat	X	X	X
Wheat and/or Corn and/or Oat, Groats and/or Barley, hulled[2]	X	X	
Brewer's dried yeast	X	X	X
Alfalfa leaf meal			X
Iodized salt	X	X	X
Vitamin A feeding oil			X
Vitamin A and D oil	X	X	
Riboflavin supplement			X
Certified food color			X
D-activated plant sterol			X
Meat by-product			X
Dried fermentation solubles	X	X	
Iron oxide (ferric oxide)	X	X	
Flavor additive to increase palability to dogs	X	X	X

[1]At least 3 percent of dried skimmed milk.
[2]At least 2 of these cereals processed or flaked shall be present, no one in an amount greater than 30 percent of the total formula.

3.3 Processing.

3.3.1 *All forms.* Dog food with cooked cereals (all classes) shall be prepared by thoroughly mixing, grinding or compressing the ingredients indicated in table II to produce a finished product that complies with 3.4. For classes 1 and 2 product the cereal ingredients shall be ground, cooked until no raw starch remains, and dried before being incorporated into the feed. Alternatively, they may consist of flaked cereals which have been heat-treated, without scorching, until no raw starch remains.

3.4 Finished product.

3.4.1 *Storage requirements.* The finished product shall not be stored more than 60 days before shipment.

TABLE III—*Sieve requirements*

Form and type (applicable to all classes)		Requirements				
		Retained on 1 inch	Pass thru No. 8	Retained on No. 4	Retained on No. 20	Pass thru No. 40
Form A,	(Meal)	—	—	4 oz.	—	12 oz.
Form B,	(½ x 1 inch pellets)	0	6 oz.	—	—	—
Form C,	(irregular pellets)	4 oz.	6 oz.	—	—	—

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TABLE IV—*Nutrients*

Nutrients	Minimum value			Maximum value		
	Class 1	Class 2	Class 3	Class 1	Class 2	Class 3
Proximate analysis:						
Crude protein	22 percent	26 percent	25 percent	—	—	—
Crude fat	5 percent	11 percent	7 percent	—	—	—
Crude fiber	—	—	—	5 percent	5 percent	5 percent
Ash	—	—	—	9 percent	9 percent	10 percent
Moisture	—	—	—	10 percent	10 percent	10 percent
Minerals and vitamins:						
Calcium	1.0 percent	1.5 percent	1.0 percent	—	—	—
Phosphorus	0.8 percent	1.2 percent	0.8 percent	—	—	—
Iron	0.01 percent	0.01 percent	0.010 percent	—	—	—
Copper	—	—	0.001 percent	—	—	—
Vitamin A	3000 USP units per pound	3000 USP units per pound	3000 USP units per pound	—	—	—
Vitamin D_2	300 USP units per pound	300 USP units per pound	300 USP units per pound	—	—	—
Thiamin	2.0 mg./pound	2.0 mg./pound	2.0 mg./pound	—	—	—
Riboflavin	2.0 mg./pound	2.0 mg./pound	2.0 mg./pound	—	—	—
Niacin	18.0 mg./pound	18.0 mg./pound	18.0 mg./pound	—	—	—
Pantothenic	4.5 mg./pound	4.5 mg./pound	4.0 mg./pound	—	—	—
Pyridoxine	—	—	2.0 mg./pound	—	—	—

3.4.2 *Physical requirements.* A 5-pound unit of the finished product shall comply with the sieving requirements of table III. Sieve sizes refer to U. S. fine and coarse series designations. In addition, the finished product shall be free of bone particles greater than 1/4 inch in any dimension, mold, hair, rust, dirt, wood, paint, and metal.

3.4.3 *Analytical requirements.* The finished product shall comply with the analytical requirements of table IV.

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform to the applicable provisions of the Federal Food, Drug, and Cosmetic Act.

3.6 Workmanship. Dry feed for dogs shall be processed and stored under sanitary conditions in accordance with good commercial practice. Particular care should be taken to keep the ingredients or the feed away from insecticides or from other materials which will leave residues, in or affect the flavor of the ingredients or the feed.

3.7 Age of ingredients and storage of the feed. The ingredients, except for the iron oxide, shall not be more than 18 months old at the time of manufacture of the feed, and in accordance with good commercial practice shall be as fresh as possible without payment of premium. The feed shall not be stored for more than 60 days before shipment to the purchasing agency.

3.8 Limitation on quantity in a single lot. The procuring agency shall be permitted to receive deliveries of the mixed feed in such quantity that it may be fed in a limited time to insure freshness. The minimum size lot shall be 100 pounds.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling. Samples of finished products covered by this specification and samples of any materials, components, or constituents entering into the preparation of the finished products may be taken at any

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time by the Government inspector to determine compliance with the requirements.

4.2 Inspection.

4.2.1 *General.* Inspection shall be made by the procuring agency or a duly authorized representative, at the time and place designated by the procuring agency. It may be made at the site of manufacture, both during and after processing, at any suitable point in transit, and after delivery to the point of destination.

4.2.2 *Factory inspection.* The passing as satisfactory of any detail of processing or materials shall not relieve the supplier of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance.

4.2.3 *Final inspection.* Unless otherwise specified (see 6.1) final inspection shall be made after delivery to the initial point of domestic destination.

4.2.4 *Quality assurance provisions.*

4.2.4.1 *Responsibility for inspection.* The supplier is responsible for meeting all the provisions of the specification prior to submission of supplies to the Government for acceptance. Unless otherwise specified (see 6.1) the supplier is required to perform the examination and tests prescribed herein, using his own or any other inspection facilities and services acceptable to the Government.

4.2.4.1.1 Inspection records of the examination and tests shall be complete and made available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2.4.2 *Sampling for inspection.* Sampling for inspection shall be performed in accordance with MIL-STD-105 except as otherwise indicated herein.

4.2.4.3 *Inspection of components.* Examination and testing of components shall be in accordance with 4.2.4.3.1 through 4.2.4.3.5 and subsidiary specification to the extent applicable, except that this specification shall govern in the event of conflict.

4.2.4.3.1 *Examination for age.* Compliance with the requirements of 3.2 for all ingredients shall be determined by a review of labels, invoices and similar documents. Failure to comply with the above requirements shall be cause for rejection of the involved lot or the product made using that lot.

4.2.4.3.2 *Examination of items 1 through 16 of table I.* Compliance with the analytical requirements of items 1 through 16 of table I shall be determined by a review of labels, shipping and storage records and similar documents. In addition a 1/2 pound sample shall be drawn from each of 5 primary containers (or all containers if 5 or less form a lot) and examined to determine compliance with physical requirements. Finding of any component which does not comply with the analytical requirements referenced above or which is not sound, wholesome and free from sour and musty odors and mold shall be cause for rejection of the involved component lot or the product made using that lot.

4.2.4.3.3 *Examination of salt.* Examination of salt shall be in accordance with that portion of the quality assurance provisions of SS-S-31 which refers to appearance and palatability. In addition, a review of labels, invoices and similar documents shall be the basis for determining if the salt is iodized. Failure to determine that the salt has been iodized shall be cause for rejection of the involved lot of salt or the product made using that lot.

4.2.4.3.4 *Examination of corn and wheat.* Compliance with the grade requirement of items 18 and 19 of table I shall be based up-

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on USDA grade certificate. Failure to furnish the above referenced certificate shall be cause for rejection of the involved component lot or the product made using that lot.

4.2.4.3.5 *Testing of animal fat.* Testing of animal fat for iodine value shall be in accordance with 4.3.1. The sample for testing shall be a 1 pound composite derived from 5 primary containers (or all containers if 5 or less form a lot). Results shall be reported to the nearest iodine value. Results less than 48 or more than 60 shall be cause for rejection of the involved component lot or the product made from that lot.

4.2.4.4 *Processing inspection.* Examination shall be made to determine compliance with the processing requirements of 3.3.1 and 3.3.2. Failure to comply with the above requirements shall be cause for rejection of the involved lot.

4.2.4.5 *Inspection of the finished product.* Examination and testing of the finished product shall be in accordance with 4.2.4.5.1 through 4.2.4.5.4.

4.2.4.5.1 *Examination of preparation for delivery.* When shipping containers are required to be in accordance with UU-S-48, examination of the filled and sealed shipping sack shall be in accordance with the external examination of the end items (except as pertains to count) as detailed in said specification. The sample size shall be as indicated by inspection level S-4 and the acceptable quality level for defects shall be 4.0 expressed as defects per hundred units, for each table of examination.

4.2.4.5.2 *Examination for sieving requirements.* The sample for examination of sieve requirements shall be a 5-pound composite derived from the number of primary containers indicated by inspection level S-2. Lot size shall be expressed in terms of primary containers. Failure to comply with all applicable requirements of table III shall be cause for rejection of the lot.

4.2.4.5.3 *Examination for prohibitive material.* Lot size shall be expressed in terms of primary containers, the sample unit shall be the contents of 1 primary container, and the sample size for inspection shall be the number of sample units indicated by inspection level S-2. Finding bone particles greater than 1/4 inch in any dimension, mold, hair, rust, dirt, wood, paint or metal shall be cause for rejection of the lot.

4.2.4.5.4 *Testing of finished product.* Testing of the finished product for the requirements of table IV shall be in accordance with 4.3.1. The sample for testing shall be a 5-pound composite derived from the number of primary containers indicated by inspection level S-2. Lot size shall be expressed in terms of primary containers. Results shall be reported to the nearest decimal specified for the respective requirement. Failure to comply with the requirements of table IV shall be cause for rejection of the lot.

4.3 **Tests.** Test procedures and controls which differ from those specified herein, may be used by the contractor if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide, as a minimum, such quality assurance, the contractor will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.3.1 *Chemical and biological tests.* Unless otherwise specified, tests shall be made in accordance with the applicable method of either the Pharmacopeia of the United States or the Official Methods of Analysis of the Association of Official Agricultural Chemists (see 6.1).

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

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5.1 Packaging.

5.1.1 *Level C.* Unless otherwise specified (see 6.1), commercial packages are acceptable.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Levels A and B.* The product shall be packed in a multiwall paper shipping sack conforming to the requirements in UU-S-48, table entitled "Commodities and their minimum sack requirements" for dog food, dry.

5.2.2 *Level C.* The feed shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Labeling and marking.

5.3.1 *Civil agencies.*

5.3.1.1 *Labeling.* Any commercial labeling, and additional labeling as specified (see 6.1) that complies with the Federal Food, Drug, and Cosmetic Act, and regulations promulgated thereunder, is acceptable. The crude protein, fat, fiber, ash, and moisture content of the manufactured feed shall be stated, either on the container or on a suitable tag attached thereto.

5.3.1.2 *Marking for shipment.* Shipping containers shall be marked in accordance with the requirements of Fed. Std. No. 123, and such other information as specified by individual agencies (see 6.1).

5.3.2 *Military agencies.*

5.3.2.1 *Labeling.* Labeling in accordance with 5.3.1.1 will be acceptable.

5.3.2.2 *Marking of shipping containers.* Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Class and form of product required (see 1.2).
(*c*) Whether preaward sample is to be furnished (see 3.1).
(*d*) When final inspection is to be made other than after delivery to the initial point of domestic destination (see 4.2.3).
(*e*) When other than the supplier is to perform the examinations and tests (see 4.2.4.1).
(*f*) When chemical and biological tests are required (see 4.3.1).
(*g*) When other than commercial packaging is required (see 5.1.1).
(*h*) Level of packing required (see 5.2).
(*i*) Additional labeling required (see 5.3.1.1).
(*j*) Additional marking required (see 5.3.1.2).

6.2 If samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 Contractor's inspection. The contractor should be required to assure that the product conforms with the specification prior to submission to the Government for final inspection.

6.4 Destination inspection. When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quantity only.

6.5 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

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Rail:

Feed, animal, prepared, not otherwise indexed by name.
Carload minimum weight 40,000 pounds.

Motor:

Feed, animal, prepared, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—GL

Navy—YD

Air Force—69

Review interest:

Army—MD

Navy—YD

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User:

Army—SM

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